



LSTM-based QoS prediction for 5G-enabled Connected and Automated Mobility applications

≡ Authors	Sokratis Barmounakis, et al
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👤 Assignee	👤 George Kyriakopoulos
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Summary

- **QoS Prediction:** LSTM-based for 5G-enabled CAM applications, focusing on Tele-operated Driving (ToD), addressing the limited studies on Deep Learning-based solutions for V2X communications and CAM
- **Prediction Scheme:** LSTM-based Multivariate Multi-step Autoencoder architecture to predict future uplink throughputs with evaluation using the NS-3 simulator on an Urban Macro deployment
- **Results:** Promising overall NMSE and RMSE, as well as 75% absolute error values under 2 Mbps for 7th second predictions
- **Future work:** Enhancing ML framework, optimizing for CAM-specific data, extending predicted QoS metrics, larger-scale scenarios
- **Keywords:** Quality of Service, Quality of Service Prediction, LSTM, RNN

Abstract

- The paper utilizes Machine Learning, Recurrent Neural Networks (RNNs), particularly LSTMs, to predict Quality of Service (QoS) changes for Connected and Automated Mobility (CAM) applications in 5G networks
- It introduces a novel QoS prediction mechanism based on an LSTM architecture and network-related QoS metrics to anticipate future QoS for Connected and Automated Vehicles (CAVs), aiming to enable proactive adaptation

Introduction

- The significance of Vehicle-to-Everything (V2X) communications for future Intelligent Transportation Systems (ITS) and Connected and Automated Mobility (CAM) applications is introduced, highlighting the strict Quality of Service (QoS) requirements
- Discussion of enhancements in the 5G System (5GS), including both the 5G Core (5GC) and the 5G Radio Access Network (RAN), to address V2X communication-related challenges and requirements.
- The role of Machine Learning (ML) and Artificial Intelligence (AI) in the intelligentization of mobile and wireless systems is highlighted, particularly in addressing network management aspects and improving reliability and QoS for CAM and V2X communications.
- The concept of predictive QoS is introduced and a QoS prediction scheme is proposed, focusing on the Tele-operated Driving (ToD) use case, aiming to proactively adapt V2X applications based on predicted QoS changes

Related Work

- Existing works on QoS prediction for 5G networks and V2X communications are limited, with few studies focusing on the specific requirements of Connected and Automated Mobility (CAM) applications
- Previous approaches include using Neural Networks (NN), auto-regressive integrated moving average (ARIMA) models, and combination methods like LSTM with k-medoids clustering for QoS prediction

- Comparison between approaches shows variations in prediction accuracy and computational efficiency, with some models struggling with scalability and complex scenarios
- The proposed work seeks to address the gap in Deep Learning-based QoS prediction for V2X communications and CAM applications, focusing on LSTM architecture for fine-grained time series prediction

QoS Prediction Scheme

- The proposed QoS prediction scheme utilizes an LSTM-based Multivariate Multi-step Autoencoder architecture to predict the uplink throughput of a ToD vehicle
- It considers various service and network-related features, including the vehicle's location, speed, distance from the cell, cell load percentage, number of connected vehicles in the cell, and uplink throughput
- The scheme employs an Encoder-Decoder LSTM architecture for sequence-to-sequence prediction, with the encoder processing input sequences and encoding them into fixed-length vectors, and the decoder reconstructing the sequence and outputting predicted future uplink throughputs

Evaluation

- The evaluation of the LSTM-based QoS prediction scheme utilized data generated using the discrete-event network simulator NS-3, on a simulated Urban Macro deployment for connected cars known as the "Manhattan Grid," with varying traffic load levels and 200 ms sampling frequency
- The LSTM model's batch size was set to 64, with a total of 500 epochs, utilizing the Adam optimizer with a learning rate of 0.0001, and using a 21 second historical time window and a 7 second prediction time window
- Performance evaluation metrics included Normalized MSE (NMSE) and Root MSE (RMSE), with the LSTM model exhibiting an overall NMSE of 0.0539 and an overall RMSE of 2.458 Mbps
- The model demonstrated successful prediction of uplink throughput fluctuations for the ToD-UE, with the 75% of absolute error values falling

below 1.832 and 1.874 Mbps for overall and 7th-second predictions, respectively

Conclusion and Future Work

- Investigated the applicability and validity of an LSTM-based QoS prediction scheme for processing ToD application, and network-oriented data to predict uplink throughput QoS metric for future time horizons
- Described the architecture of the LSTM-based mechanism and presented promising evaluation results for the specific use case
- Future steps include further investigation and enhancement of the Machine Learning framework, optimization for CAM-specific application data, extension of predicted QoS metrics, and experimentation with larger-scale scenarios for higher time horizons